KNOWLEDGE SYSTEMS WORKING GROUP

Strategic Proposal

Enabling Semantic Interoperability Across

The INCOSE Ecosystem

Strategic Value Proposition

Knowledge Systems WG serves as the central semantic integrator for 50+ INCOSE working groups, delivering annual impact and ongoing value through enhanced understanding, communication, coordination, and impact.

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Executive Summary

The Opportunity

INCOSE has cultivated a rich ecosystem of 50+ working groups with deep specialized expertise spanning domains from aerospace to healthcare, methodologies from MBSE to agile systems engineering, and foundational topics from systems science to human factors integration. This represents an extraordinary concentration of systems engineering knowledge.

The Challenge

Despite this wealth of expertise, working groups operate with incompatible terminologies, conceptual models, and knowledge structures. This fragmentation creates significant barriers:

- Cross-group collaboration is hindered by semantic gaps
- Knowledge integration opportunities are missed
- Systemic insights requiring multiple domains cannot be synthesized
- AI/ML systems cannot effectively leverage distributed INCOSE knowledge
- Redundant efforts in ontology and framework development

The Solution

The Knowledge Systems Working Group is uniquely positioned to serve as the semantic integrator across the entire INCOSE ecosystem by providing:

Value Type	Description
Understanding	Formal ontologies and category theory frameworks provide shared conceptual models enabling groups to understand each other's domains
Communication	Unified semantic layer and shared vocabularies enable unambiguous communication across 50+ working groups
Coordination	Enterprise knowledge graphs and semantic governance provide infrastructure for coordinated cross-group initiatives
Impact	RAG 2.0 and the Dynamic Causal Factor Model frameworks enable AI-enhanced decision-making and multi-order impact assessment

Business Case at a Glance

This proposal delivers compelling impacts:

Metric	Impact
Annual Value Created	Cost reduction + new capabilities
Total Investment (24 months)	Infrastructure + 5 FTE team
First-Year ROI	Value exceeds investment 4.8x
Payback Period	Rapid value realization
Working Groups Enabled	Ecosystem-wide transformation

Strategic Imperative

This is not an incremental improvement—it represents a fundamental transformation of how INCOSE working groups collaborate, innovate, and deliver value. The Knowledge Systems WG has the unique opportunity to position INCOSE as the global leader in semantic systems engineering while enabling next-generation AI-augmented engineering capabilities.

Current State Analysis

INCOSE Working Group Ecosystem

INCOSE has established 53 active working groups organized across multiple dimensions:

Domain Verticals (15 Working Groups)

Industry-specific systems engineering practice areas:

- Automotive, Healthcare, Space Systems, Defense Systems, Transportation
- Power & Energy Systems, Smart Cities Initiative, Industrial Systems
- Information Communications Technology, Infrastructure

Practice Areas (18 Working Groups)

Systems engineering methodologies and approaches:

- MBSE Initiative, MBSE Patterns, Architecture, Requirements Engineering
- Integration, Verification & Validation, Configuration Management
- Risk Management, System Safety, Systems Security Engineering
- Quality Management (SEQM), Measurement, Agile Systems Engineering

Foundational Topics (12 Working Groups)

Core systems thinking and theoretical frameworks:

- Systems Science, Complex Systems, System of Systems
- Human Systems Integration, Social Systems, Natural Systems
- Decision Analysis, Resilient Systems, Sustainability

Professional Development (8 Working Groups)

Competency development and organizational transformation:

- Competency, SE in Academia, SE in Early Stage R&D
- Embedding Systems Engineering Into Organizations
- Empowering Women Leaders in SE, Small Business Systems Engineering

Knowledge Fragmentation Challenges

Despite this organizational structure, significant challenges limit cross-group collaboration and knowledge integration:

Challenge 1: Semantic Gaps

Working groups use incompatible terminologies for the same concepts. For example, 'requirement' means different things in the Requirements WG, the MBSE Initiative, and the Healthcare WG. Without shared ontologies, communication breaks down.

Challenge 2: Redundant Development

Multiple working groups independently develop overlapping frameworks. Architecture WG, Enterprise SE WG, and MBSE Initiative all create metamodels without coordination, wasting duplicate effort across working groups annually.

Challenge 3: Missed Integration Opportunities

Critical cross-domain insights require expertise from multiple working groups. For example, sustainable smart cities require integration of the Sustainability WG, Smart Cities WG, Measurement WG, and Social Systems WG—but no infrastructure exists to enable this collaboration.

Challenge 4: AI Systems Cannot Leverage INCOSE Knowledge

AI/ML systems trained on INCOSE content struggle with inconsistent terminology and conceptual models. Without semantic grounding, AI systems hallucinate incorrect answers when asked about systems engineering topics, limiting the adoption of AI-augmented tools.

Future State Proposal

Establish KSWG serves as the Semantic Integrator

The Knowledge Systems Working Group is uniquely positioned to serve as the central semantic integrator for the entire INCOSE ecosystem. Unlike domain-specific working groups, KSWG provides foundational infrastructure that enables all other groups to:

- Share common conceptual frameworks through formal ontologies
- Communicate unambiguously through standardized vocabularies
- Coordinate initiatives through enterprise knowledge graphs
- Amplify impact through AI-enhanced decision support

Architecture Overview

The proposed solution consists of four integrated architectural layers:

Layer 1: Ontology Infrastructure (Understanding)

Formal semantic models provide shared understanding across all working groups.

Component	Description
Upper Ontology	Foundational concepts shared across all domains: Entity, Process, Relationship, System, Component, Property. Provides a common vocabulary for all working groups.
Core Ontologies	Mid-level concepts bridging general and specific: Requirement, Architecture, Risk, Stakeholder, Interface, Verification. Enables cross-domain integration.
Domain Ontologies	Specialized models for each working group: Healthcare Clinical Ontology, Automotive Supply Chain Ontology, Space Mission Ontology. Total of 50+ domain-specific models.
Category Theory	Mathematical validation of cross-domain transformations. Ensures structural preservation when mapping concepts between domains (e.g., healthcare requirements → medical device architecture).

Layer 2: Semantic Knowledge Layer (Communication)

Unified infrastructure enabling unambiguous communication across working groups.

Component	Description
Unified Semantic Layer	Enterprise-wide metadata harmonization enabling business users to query data using domain concepts rather than technical schemas. Three-tier architecture (Presentation/Application/Data) supporting data literacy initiatives.
SKOS Vocabularies	Standardized terminologies using Simple Knowledge Organization System (SKOS) with multilingual support. Lightweight taxonomies providing consistent naming across working groups without requiring full ontology adoption.
RDF/OWL Standards	Machine-readable semantic models using Resource Description Framework (RDF) and Web Ontology Language (OWL). Enables automated reasoning, inference, and validation of knowledge consistency.
JSON-LD Integration	Web-compatible semantic data exchange using JSON for Linking Data (JSON-LD). Bridges traditional web APIs with semantic technologies, enabling tool integration without requiring RDF expertise.

Layer 3: Knowledge Graph Infrastructure (Coordination)

Connected semantic data structures enabling coordinated cross-group initiatives.

Component	Description
Enterprise Knowledge Graph	Connected expertise across 50+ domains stored in a graph database (GraphDB/Neo4j). Enables discovery of relationships and patterns invisible in siloed data. Target: 1M+ triples covering all INCOSE working groups.
Semantic Governance	Management framework defining metadata standards, approval workflows for ontology changes, and quality metrics. Implemented change control processes with subject matter expert review and formal tracking/monitoring.
Traceability Infrastructure	End-to-end relationship tracking from stakeholder needs through requirements, design,

	implementation, and verification. Enables impact analysis showing ripple effects of changes across multiple working group domains.
SPARQL Query Interface	Powerful query language enabling complex knowledge discovery. Working groups can execute sophisticated queries like 'Find all safety requirements affecting both space systems and human factors across autonomous vehicle projects'.

Layer 4: AI Enhancement Layer (Impact)

Artificial intelligence capabilities are amplifying decision-making and impact assessment.

Component	Description
RAG 2.0 Architecture	Knowledge graph-enhanced Retrieval-Augmented Generation for AI systems. Pilot results show 40% improvement in AI accuracy by grounding responses in verified knowledge graphs rather than probabilistic text generation alone.
DCFM Framework	Dynamic Causal Factor Model enabling 'Do No Harm' evaluation framework. Evaluates decisions through a multi-stakeholder lens, analyzing first-order, second-order, and systemic effects across all impacted working groups.
Chaos Theory Modeling	Nonlinear dynamics and emergent behavior prediction enabling Complex Systems WG to model tipping points, strange attractors, and bifurcations in socio-technical systems. Complements traditional linear modeling approaches.
Semantic Search	Meaning-based search understands conceptual relationships rather than just keyword matching. For example, searching 'vehicle safety' automatically includes results for 'automotive hazard mitigation' and 'transportation risk management'.

Cross-Group Integration Examples

The following examples demonstrate how Knowledge Systems WG enables high-value collaboration across working groups:

Example 1: Digital Engineering Cluster

Participants: MBSE Initiative + AI Systems + Digital Engineering Information Exchange + Tools Integration

The Challenge

AI systems cannot reason effectively over SysML models because models lack formal semantics. Different tools use incompatible data formats, preventing model exchange. Design space exploration requires manual review of thousands of model variants.

Knowledge Systems Solution

KSWG provides a formal SysML ontology enabling AI to understand model elements and relationships. Semantic data exchange layer enables tool-to-tool integration. RAG 2.0 architecture allows AI to validate models against requirements and suggest optimizations automatically.

Measurable Impact

50% reduction in model review time through AI-assisted validation. Automated consistency checking catches 85% of common modeling errors. Design space exploration evaluates 10x more alternatives through AI optimization.

Example 2: Assurance Cluster

Participants: Decision Analysis + Risk Management + System Safety + Systems Security Engineering

The Challenge

Decision analysis, risk assessment, safety analysis, and security analysis are performed in isolated silos using incompatible tools and frameworks. Decisions optimizing for cost may introduce safety hazards. Security controls may create operational risks. No integrated view exists.

Knowledge Systems Solution

KSWG integrates decision, risk, safety, and security ontologies into a unified assurance framework using DCFM. Knowledge graph traces relationships across all four domains. Multi-order impact analysis evaluates second-order and systemic effects of decisions across stakeholder groups.

Measurable Impact

30% reduction in safety incidents through integrated analysis, catching cross-domain hazards. Decision confidence scores improve 45% when stakeholder impacts are quantified. The cost of rework from incompatible analyses is greatly reduced annually.

Example 3: Impact Cluster

Participants: Sustainability + Measurement + Social Systems + Natural Systems

The Challenge

Sustainability claims lack quantifiable, standardized metrics. Environmental impact assessments don't include social equity dimensions. Natural systems expertise is isolated from urban planning. No triple bottom line reporting framework exists.

Knowledge Systems Solution

KSWG provides an integrated impact ontology covering environmental, social, and economic dimensions. Metrics ontology enables standardized sustainability KPIs. Knowledge graph connects ecosystem models with urban systems and social equity frameworks.

Measurable Impact

ESG reporting compliance achieved for 90% of participating organizations. Sustainability metrics become auditable with full data lineage. Social equity considerations are integrated into 100% of infrastructure projects. Investor confidence increases measurably.

Business Case & Financial Analysis

Quantifiable Value Creation

The Knowledge Systems WG initiative delivers measurable impact and efficiencies:

Cost Reduction:

Cost Reduction Category	Basis
Eliminate Redundant Ontology Development	10 groups × $50K each
Reduce Integration Effort	20% reduction in cross-group projects
Accelerate Decision-Making	DCFM reduces analysis time 30%
Total Annual Cost Savings	

Value Creation

Value Creation Category	Basis
Accelerate Innovation	Faster cross-domain insights
Improve Quality	Better traceability, fewer defects
Enable AI Capabilities	RAG 2.0 enabling new applications
Total Annual Value Creation	

Implementation Roadmap

Phase 1: Foundation (Months 1-6)

Objective: Establish KSWG as the recognized semantic integrator

Key Activities:

- Publish Knowledge Systems Primer (per IW26 commitment)
- Develop Upper Ontology adopted across 5 working groups
- Establish a Semantic Governance Framework as an INCOSE standard
- Pilot RAG 2.0 with AI Systems WG

Success Metrics:

- 5 working groups adopt Upper Ontology
- 10 working groups reference KSWG in deliverables
- RAG 2.0 pilot shows >30% accuracy improvement

Investment: 2 FTE semantic engineers + infrastructure

Phase 2: Expansion (Months 7-12)

Objective: Scale semantic integration across 20 working groups

Key Activities:

- Develop 20 Domain Ontologies (one per major working group)
- Deploy Enterprise Knowledge Graph (pilot with 5 working groups)
- Establish Integration Clusters (Digital Engineering, Assurance, Impact, Theory)
- Publish DCFM Framework as an INCOSE guide

Success Metrics:

- 20 working groups contribute domain knowledge
- Knowledge graph contains 10,000+ concepts
- 3 integration clusters actively collaborating
- 5 working groups using DCFM for decisions

Investment: 3 FTE + knowledge graph deployment

Phase 3: Maturity (Months 13-24)

Objective: Achieve ecosystem-wide semantic interoperability

Key Activities:

- Complete 50+ Domain Ontologies covering all working groups
- Deploy Production Knowledge Graph (enterprise-scale)
- Integrate RAG 2.0 with INCOSE Digital Library
- Establish KSWG as the INCOSE Center of Excellence

Success Metrics:

- All 50+ working groups are integrated
- Knowledge graph queries average <100ms
- RAG 2.0 serving 1000+ users
- 90% working groups cite the KSWG value

Investment: 5 FTE + production operations

Conclusion and Call to Action

The Knowledge Systems Working Group has a unique and strategic opportunity to serve as the semantic integrator for the entire INCOSE ecosystem. This proposal demonstrates:

- Clear value proposition across four dimensions: Understanding, Communication, Coordination, and Impact
- Compelling impact for INCOSE (short term) and SE practitioners (long term)
- Concrete integration examples delivering measurable benefits across working group clusters
- Phased implementation roadmap with clear success metrics at each stage
- Strategic positioning as a global leader in semantic systems engineering

The Strategic Question

This is not an incremental improvement—it represents a fundamental transformation of how INCOSE working groups collaborate, innovate, and deliver value. The question is not whether Knowledge Systems should play this role, but how quickly we can deploy this capability across the INCOSE ecosystem.

We respectfully request approval to proceed with Phase 1: Foundation, investing in a 6-month effort to establish KSWG as the recognized semantic integrator for INCOSE.

For questions or additional information, please contact:

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